

ZADOK MBITI

INFORMATION SYSTEMS • DATABASE MANAGEMENT • ICT PROJECT & SECURITY MANAGEMENT

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PROFESSIONAL PROFILE

ICT Management graduate (coursework complete; graduating 13 November 2026) with a track record of building and running production-grade, database-driven information systems end to end, from relational data modelling and secure authentication through to deployment, documentation and client handover. Academic performance, Scored A grades across Network Security Management, Systems Support, Organization Theory, Object-Oriented Analysis & Design, Workflow Applications and Software Quality Assurance. Strongest in information systems strategy, database design, IT security/access control, systems administration and IT project management, backed by hands-on cloud and infrastructure experience gained at KoTDA, an attachment graded A. Comfortable owning a system from requirements through to a live client-facing product.

CORE STRENGTHS

Information Systems & Databases: Relational data modelling, PostgreSQL & MySQL, advanced & emerging database technologies, data integrity (constraints, indexing, validation), operational reporting and analytics.

IT & Security Management: Security policy, network security management, access control, authentication & session security, IT service cost/availability management, IT investment planning.

IT Project & Business Management: IT project management, information systems strategy, business intelligence & decision support systems, requirements gathering, technical documentation, stakeholder/people management.

Systems Administration & Infrastructure: Linux/Windows administration, network server maintenance, cloud provisioning (AWS EC2), Nginx reverse proxying & load balancing, backup and monitoring support.

Software Development: Python (FastAPI, Django, Flask), PHP, JavaScript, REST API design, M-Pesa Daraja & SMS/USSD integration, scored grade A in Software Quality Assurance.

PROFESSIONAL EXPERIENCE

IT / Cloud Computing Intern | Konza Technopolis Development Authority (KoTDA) May 2025 – Aug 2025
Formal industrial attachment component of the BSc program graded A.

- Supported ICT systems and infrastructure operations within the Cloud team, Department of ICT and Smart City Solutions.
- Provisioned and configured virtual machines and cloud environments, including Ubuntu-based systems and AWS EC2 resources.
- Supported LAN/WAN infrastructure, TCP/IP networking, system monitoring and troubleshooting.

- Assisted with backup procedures, access controls and security-related ICT operations.
- Prepared and maintained technical documentation covering ICT processes, configurations and operational procedures.
- Supported deployment and administration of web and application services in Linux and Windows environments.

KEY PROJECTS

Drumvale Riverview Welfare Group - Welfare/Chama Management Platform *FastAPI, PostgreSQL (Neon), JavaScript, Vercel*

- Designed and built a live, production full-stack platform (drumvalewelfare.com) managing member records, contributions, welfare cases and multi-role access (super admin, admin, secretary, chairperson, treasurer, member).
- Implemented JWT authentication with OTP-based password reset, DB-backed rate limiting, login logout and privilege-escalation fixes as part of a dedicated security-hardening pass.
- Integrated Africa's Talking SMS and M-Pesa Paybill tracking for contribution and meeting notifications; migrated the database from Render to Neon PostgreSQL with zero data loss.
- Produced client-facing documentation, a user manual, presentation deck and security booklet and manage the platform's custom domain and email infrastructure.

SmartStock AI - POS, Inventory & Demand Forecasting Platform *FastAPI/TypeScript, PostgreSQL, Prophet, React*

- Built an inventory and POS system recording sales, stock movements, suppliers, purchase orders and payment references, with an auditable stock-movement model.
- Implemented structured analytics (revenue, gross profit, low-stock counts, top sellers) and explainable demand forecasting with reorder recommendations and stockout-risk reporting.

E-Commerce & Information Management Platform (Jewellery Retailer) *FastAPI, PostgreSQL, HTML/CSS/JavaScript*

- Designed relational database structures (foreign keys, indexes, unique constraints, validation) to maintain data integrity across products, customers, orders and reviews.
- Built authenticated customer accounts, order-history and admin workflows, plus security controls including password hashing, rate limiting and temporary IP logout.

Reverse Proxy & Load-Balancer Infrastructure *Nginx, FastAPI, Redis, Kali Linux/WSL*

- Configured a Nginx reverse proxy with round-robin load balancing across two FastAPI instances managed via NSSM.
- Integrated Redis for shared session storage and JWT token blacklisting across instances.

Additional: FreshWash (Django laundry management) & Zaddy Business Directory (PHP/MySQL) *M-Pesa Daraja, Africa's Talking*

- FreshWash full-stack order intake, staff and payment workflows with M-Pesa STK Push and SMS/USSD access, deployed on Railway.
- Zaddy Business Directory - Kenyan business-listing platform with subscription billing via M-Pesa Daraja STK Push and category/keyword search.

EDUCATION

Bachelor of Science, Information and Communication Technology Management

Maseno University, Kenya Coursework complete; graduating 13 November 2026 • A grades in Network Security Management, Systems Support, Organization Theory, Object-Oriented Analysis & Design, Workflow Applications, Software Quality Assurance and Industrial Attachment

- **Information Systems & Databases:** Database Systems, Advanced Database Systems, Emerging Database Technologies & Applications, Data Structures & Algorithms, Object-Oriented Analysis & Design (A).
- **IT Security & Administration:** Security Policies, Network Security Management (A), Systems Support (A), System Administration I & II, Network Server Maintenance, IT Hardware Support.
- **IT & Business Management:** IT Project Management, Information Systems Strategy, Managing the IT Investment, IT Service Cost & Availability Management, Business Intelligence, Decision Support Systems, Management of Information Systems, Organization Theory (A), Entrepreneurship & Small Business Management, People Management.
- **Software Engineering:** Software Engineering, Software Quality Assurance (A), Object-Oriented Programming I & II, Web Application Client Programming, Systems Analysis & Design, Workflow Applications (A).
- **Applied Project Work:** Industrial Attachment at KoTDA (A); IT Individual Project Proposal Writing & Implementation; Group ICTM Project Proposal Writing & Implementation.

TECHNICAL SKILLS

- **Operating Systems & Networking:** Linux/Ubuntu, Windows/IIS, TCP/IP, LAN/WAN, network troubleshooting and monitoring, pfSense, Proxmox.
- **Systems Administration & Cloud:** AWS EC2, Nginx, Apache2, Redis, Docker/Kubernetes fundamentals, Railway, Vercel, backup and access-control support.
- **Databases & Data:** PostgreSQL, MySQL, SQL, Drizzle ORM, relational data modelling, operational analytics and reporting.
- **Programming & APIs:** Python, Django, FastAPI, PHP, Node.js/Express, JavaScript, REST APIs, OpenAPI.
- **Web & Frontend:** HTML5, CSS3, React, Expo/React Native, Next.js/TypeScript (in progress).
- **Security & Integrations:** Authentication & access control, M-Pesa Daraja API, Africa's Talking SMS/USSD.
- **Tools & Productivity:** Git/GitHub, VS Code, XAMPP, Microsoft Office.

ADDITIONAL STRENGTHS

- Fast learner who adapts quickly across ICT domains from systems administration to software development to information security.
- Strong technical documentation, troubleshooting and systems-support capability, developed both academically and at KoTDA.
- Analytical approach to information reporting, forecasting, audit trails and data-quality assurance.
- Proven ability to translate operational and business requirements into secure, database-backed information systems delivered to real clients.
- Comfortable working independently or within a team, with practical experience communicating technical work to non-technical stakeholders.

REFERENCES

Available upon request.